ADAPTIQ — ADAPTIVE INTELLIGENCE FOR PERSONALIZED LEARNING

[CS-401] DISTRIBUTED SYSTEMS & CLOUD COMPUTING

Category: Computer Science • Generated for: sibigathan • Date: Oct 14, 2026, 11:14 PM
Grounding: 100% Derived from 1 Verified Course Document(s)

PART A: 2-MARK QUESTIONS & VERIFIED ANSWERS

Concise Definitions, Key Invariants, Operational Differences & Formulas

Total Questions: 7 • Marks per Question: 2 Marks • Total Marks: 14 Marks

INDEX OF COVERED TOPICS & LEARNING UNITS:

1. Byzantine Fault Tolerance • 2. Logical Clocks • 3. Raft Consensus Protocol • 4. Paxos State Machine Replication • 5. CAP Theorem • 6. DynamoDB Gossip Protocol

QUESTION 1 Topic: Byzantine Fault Tolerance **[2 Marks]**

Q1. State the FLP impossibility theorem regarding consensus in distributed systems.

Difficulty: HARD | Bloom: Remember

[SOURCE GROUNDING]: Distributed Systems Principles Textbook (Page 1) — "Lamport proved that no agreement can be reached in an asynchronous network with even one crash failure under t..."

VERIFIED MODEL ANSWER & KEY INVARIANTS

Lamport proved that no agreement can be reached in an asynchronous network with even one crash failure under the FLP impossibility theorem. This fundamental theorem highlights the limits of fault tolerance in asynchronous distributed systems.

QUESTION 2 Topic: Logical Clocks **[2 Marks]**

Q2. How do Lamport Logical Clocks update timestamps when receiving an event from another process?

Difficulty: MEDIUM | Bloom: Understand

[SOURCE GROUNDING]: Distributed Systems Principles Textbook (Page 2) — "Lamport Logical Clocks assign a monotonically increasing integer timestamp C(e) to every event. If event a hap..."

VERIFIED MODEL ANSWER & KEY INVARIANTS

Lamport Logical Clocks assign a monotonically increasing integer timestamp C(e) to every event. When a message is received, the receiver updates its clock to the maximum of its current local clock and the received timestamp, plus one: max(C(receive), C(send)) + 1.

QUESTION 3 Topic: Raft Consensus Protocol **[2 Marks]**

Q3. What are the three independent subproblems into which the Raft consensus protocol decomposes consensus?

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: Distributed Systems Principles Textbook (Page 3) — "Raft decomposes consensus into three independent subproblems: Leader Election, Log Replication, and Safety...."

VERIFIED MODEL ANSWER & KEY INVARIANTS

Raft decomposes consensus into three independent subproblems: Leader Election, Log Replication, and Safety. This modular approach simplifies the design and understanding of the consensus mechanism.

QUESTION 4 Topic: Raft Consensus Protocol **[2 Marks]**

Q4. What triggers a follower node to transition to a Candidate state in the Raft protocol?

Difficulty: MEDIUM | Bloom: Understand

[SOURCE GROUNDING]: Distributed Systems Principles Textbook (Page 3) — "If a follower receives no communication over an election timeout, it transitions to Candidate and requests vot..."

VERIFIED MODEL ANSWER & KEY INVARIANTS

If a follower receives no communication over an election timeout, it transitions to the Candidate state and begins requesting votes via RequestVote RPCs.

QUESTION 5 Topic: Paxos State Machine Replication **[2 Marks]**

Q5. Name the two-phase rounds used by Classic Multi-Paxos to reach consensus across distinct log slots.

Difficulty: MEDIUM | Bloom: Remember

[SOURCE GROUNDING]: Distributed Systems Principles Textbook (Page 4) — "Classic Multi-Paxos reaches consensus across distinct log slots through two-phase rounds: Phase 1 Prepare and ..."

VERIFIED MODEL ANSWER & KEY INVARIANTS

Classic Multi-Paxos reaches consensus across distinct log slots through two distinct two-phase rounds: Phase 1 Prepare and Promise, followed by Phase 2 Accept and Accepted.

QUESTION 6 Topic: CAP Theorem **[2 Marks]**

Q6. State the Brewer CAP Theorem.

Difficulty: EASY | Bloom: Remember

[SOURCE GROUNDING]: Distributed Systems Principles Textbook (Page 5) — "Brewer CAP Theorem states that any distributed data store can simultaneously provide at most two of three guar..."

VERIFIED MODEL ANSWER & KEY INVARIANTS

Brewer's CAP Theorem states that any distributed data store can simultaneously provide at most two of three foundational guarantees: Consistency, Availability, and Partition Tolerance.

QUESTION 7 Topic: DynamoDB Gossip Protocol **[2 Marks]**

Q7. How does Amazon Dynamo implement anti-entropy background synchronization?

Difficulty: MEDIUM | Bloom: Understand

[SOURCE GROUNDING]: Distributed Systems Principles Textbook (Page 6) — "Amazon Dynamo adopts an AP model using consistent hashing, vector clocks for versioning, and anti-entropy back..."

VERIFIED MODEL ANSWER & KEY INVARIANTS

Amazon Dynamo adopts an AP model that implements anti-entropy background synchronization using Merkle trees communicated via gossip protocols, alongside consistent hashing and vector clocks for versioning.